Paul Schulte

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Education

RWTH Aachen University — Oct 2025 – Present
M.Sc. Computer Science; GPA: 1.2 (1.0 best) — Aachen, Germany
- **Focus**: Machine Learning, Optimization, Algorithms, and Graph Theory

RWTH Aachen University — Oct 2022 – Aug 2025
B.Sc. Computer Science; GPA: 1.1 with distinction (1.0 best) — Aachen, Germany
- **Dean's List**: Top 5% of cohort across all semesters; completed degree in 5 of 6 semesters
- **Bachelor Thesis**: *Action Classification Through Dynamic Scene Graph Generation from Image Sequences* (Grade: 1.0)

Aalto University — Aug 2024 – Dec 2024
Semester Abroad, Computer Science; GPA: 5.0 / 5.0 — Espoo, Finland
- **Coursework**: Competitive Programming, Distributed Algorithms, Graph Theory

Research Experience

RWTH Aachen University, Chair for Learning on Graphs — Aachen, Germany
Graduate Research Assistant — Oct 2026 – Present
- Researching Graph Foundation Models (GFMs) in Christopher Morrises group

RWTH Aachen University, Chair for Machine Learning and Inference — Aachen, Germany
Graduate Research Assistant — Sep 2025 – Aug 2026
- Developed a neuro-symbolic framework for learning lifted symbolic planning models directly from image sequences
- Designed a spatio-temporal Graph Neural Network (GNN) to infer action types and involved objects from dynamic scene graphs
- Evaluated generalization across diverse planning domains, including previously unseen environments with varying numbers of objects
- Equal-contribution first author of an oral paper at the GenPlan Workshop (ICAPS 2026)
- Hector Geffners group

Robotics Institute, Carnegie Mellon University — Pittsburgh, PA, USA
Undergraduate Research Intern (RISE Worldwide) — May 2024 – Sep 2024
- Developed an MPC framework with Control Barrier Functions for safety-critical trajectory planning
- Implemented real-time planning in JAX for CPU-based F1TENTH hardware, reducing reliance on GPU-heavy approaches
- Co-authored the USDOT/Safety21 final report *Risk-Aware Warning and Control for Interactive Traffic Safety*

Professional Experience

Siemens Logistics — Düsseldorf, Germany
Software Engineering Working Student — Feb 2023 – Nov 2024
- Designed and implemented C# features for an airport baggage handling system planning tool
- Optimized computationally intensive planning components, reducing runtime by up to 80%

Selected Projects

Math Exercise Generator | *JavaScript, HTML, Vue.js* — 2023
- Built a web-based linear algebra exercise generator with automatic solution computation

Neuroevolution of Fixed Topologies | *Python, Reinforcement Learning* — 2021
- Implemented neural networks and evolutionary algorithms from scratch
- Applied methods to OpenAI Gym environments and documented results in a technical report

Publications

Ulzhalgas Rakhman*, **Paul Schulte***, Hector Geffner. *Learning Lifted Symbolic Planning Domains from Image Sequences (VISA).* GenPlan Workshop at ICAPS 2026. Oral presentation. *Equal contribution.

Technical Skills

Programming: Python, C++, Java, C#, JavaScript
ML / Scientific Computing: JAX, NumPy, Machine Learning, Optimization
Tools: Linux, Git, Docker, LaTeX
Languages: German (native), English (C1)

Extracurricular Activities

Board member and spoke person of the student dorm

Honors & Distinctions

DAAD RISE Worldwide Scholarship (2024) | **German Mathematical Society** student member (2022) | **German Physical Society** student member (2022)